

MRI Full Course Formula Sheet

Compact formulas for exam memorization - two lectures per page

Week 1 - NMR Basics

Spin		
Angular momentum	$\mathbf{p} = m\mathbf{v}, \mathbf{J} = \mathbf{r} \times \mathbf{p}$	Rotating charge.
Magnetic moment	$\boldsymbol{\mu} = \gamma \mathbf{J}$	γ gyromagnetic ratio.
Quantization	$J_z = \hbar m, m = -1, -1, \dots, 1$	Spin projection.
Spin 1/2	$I = 1/2 \Rightarrow m = \pm 1/2$	^1H has two states.
Moment projection	$\mu_z = \gamma \hbar m = \pm(1/2)\gamma \hbar$	Along B_0 .

Energy And Resonance		
Energy	$E_m = -\mu_z B_0 = -\gamma \hbar B_{0m}$	Parallel lower for $\gamma > 0$.
Gap	$\Delta E = \gamma \hbar B_0 = \hbar \omega_0$	Energy splitting.
Boltzmann	$n_{\text{down/up}} = e^{-\Delta E/k_B T}$	Population ratio.
Polarization	$\Delta n/n \approx \Delta E/(2k_B T)$	For $\Delta E \ll k_B T$.
Net M	$M_0 = \sum \mu = \Delta n \mu_{\text{zez}}$	Longitudinal M.
Scaling	$M_0 \propto \rho_{\text{HB0/T}}$	Proton density matters.

Dynamics		
Torque	$d\boldsymbol{\mu}/dt = \gamma(\boldsymbol{\mu} \times \mathbf{B})$	Precession.
Macroscopic	$d\mathbf{M}/dt = \gamma(\mathbf{M} \times \mathbf{B})$	Same law for M.
RF	$\mathbf{B}_{1(t)} = B_1 \cos(\omega_0 t) \mathbf{e}_x = B_1 \mathbf{L} + B_R$	Excitation field.
Rotating frame	$(d\mathbf{M}/dt)_{\text{rot}} = \gamma \mathbf{M} \times \mathbf{B} - \omega_0 \times \mathbf{M} = \gamma \mathbf{M} \times \mathbf{B}_{\text{eff}}$	Use effective field.
Effective field	$\mathbf{B}_{\text{eff}} = (B_0 - \omega/\gamma) \mathbf{e}_z + B_1 \text{rote}_x$	On resonance mostly B_1 .
Flip angle	$\alpha = \gamma \int B_1 \text{,rot}(t) dt$	RF pulse area.

Relaxation		
T2*	$1/T_{2^*} = 1/T_2 + \gamma \delta B$	Field inhomogeneity.
Bloch x	$dM_x/dt = \gamma(M_y B_z - M_z B_y) - M_x/T_2$	Transverse loss.
Bloch y	$dM_y/dt = \gamma(M_z B_x - M_x B_z) - M_y/T_2$	Transverse loss.
Bloch z	$dM_z/dt = \gamma(M_x B_y - M_y B_x) - (M_z - M_0)/T_1$	Recovery.

Week 2 - Image Formation

Encoding		
Gradient	$G_i = \partial B_z / \partial i$	$i = x, y, z$.
Linear field	$\Delta B_{z(x)} = G_{xx}$	Constant slope.
Frequency	$\omega(r) = \gamma[B_{0+G} \cdot r]$	Position dependent.
Readout	$\omega(x) = \gamma G_{xx}$	Frequency encoding.
Phase	$\Phi(y) = \gamma G_{yy} T_y$	Phase encoding.
K-space	$k(t) = \gamma \int G(t) dt$	Constant G: $k = \gamma G t$.

Signal And FT		
Signal	$s(k_x, k_y) \approx \sum \sum p(x_i, y_j) e^{-j(k_{xx} x_i + k_{yy} y_j)}$	FT of object.
Image	$p(x, y) \approx \sum \sum s(k_{xp}, k_{yq}) e^{-j(k_{xp} x + k_{yq} y)}$	Inverse FT.
K spacing	$\Delta k = 2\pi / \text{FOV}$	Nyquist.
Pixel	$\Delta x = \text{FOV}_x / N_x$	Same in y.
Max k	$k_{\text{max}} = \pi / \Delta x = \pi N_x / \text{FOV}_x$	Resolution.
Readout BW	$\text{BW}_x = \gamma G_{xx} \text{FOV}_x$	Angular freq convention.
Dwell	$\Delta t = 2\pi / \text{BW}_x, T_x = N_x \Delta t$	Sampling.

Slice And Contrast		
Gradient moment	$k_i = \gamma \int G_{it}(t) dt$	Gradient area.
Slice thickness	$\Delta z = \text{BW}_{\text{RF}} / (\gamma G_z)$	RF bandwidth.
RF selectivity	$R = \text{BW}_{\text{RFTRF}}$	Longer pulse narrower BW.
PD contrast	$C \propto \rho_A \rho_B$	Long TR.
Partial sat.	$C \propto \rho_A (1 - e^{-TR/T_1 A}) \rho_B (1 - e^{-TR/T_1 B})$	T1 weighting.
Scan time	$T_{\text{scan}} \approx N_{\text{phase}} \text{TR}$	One ky line per TR.

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Week 3 - Fast And Parallel Imaging

Speed, SNR, Resolution		
Voxel	$\Delta V = \Delta x \Delta y \Delta z$	Voxel volume.
SNR scaling	$SNR \propto \Delta V \sqrt{T_{scan}} / \sqrt{BW}$	Core tradeoff.
Averages	$SNR \propto \sqrt{NSA}$	Repeated scans.
Acceleration	$T_{scan} \cdot R \approx T_{scan}/R$	Undersampling.
Aliased FOV	$FOV_{alias} = FOV/R$	Phase undersampling.

Coil Encoding		
Coil signal	$s_{\gamma}(k) = \int p(x) c_{\gamma}(x) e^{-jkx} dx$	Sensitivity cy.
Matrix model	$s = E\rho + \eta$	Encoding plus noise.
Decode	$i = Fs$	Reconstruction matrix.
SENSE	$\hat{\rho} = (E^H \Psi^{-1} E)^{-1} E^H \Psi^{-1} s$	Optimum SNR inverse.
Regularized	$\hat{\rho} = (E^H \Psi^{-1} E + \lambda I)^{-1} E^H \Psi^{-1} s$	Stabilizes ill-conditioning.

Parallel Limits		
SENSE SNR	$SNR_{SENSE} = SNR_{full} \sqrt{R \cdot g(x)}$	g-factor penalty.
g-factor	$g(x) = \sqrt{\left[\frac{E_{ii}^{HE}}{(E_{ii}^{HE})^2 - 1} \right]} \geq 1$	Noise amplification.
Good coils	Distinct $c_{\gamma}(x) \Rightarrow$ low g	Separates aliased voxels.
Failure	$R > N_{coils}$ or poor sensitivities	Underdetermined/ill-conditioned.

Week 4 - Image Contrast

Relaxation Contrast		
T1 recovery	$M_z(t) = M_{0+} [M_z(0) - M_0] e^{-t/T_1}$	Longitudinal.
After 90°	$M_z(t) = M_0 (1 - e^{-t/T_1})$	$M_z(0) = 0$.
T2 decay	$M_{xy}(t) = M_{xy(0)} e^{-t/T_2}$	Spin-spin.
T2* decay	$M_{xy}(t) = M_{xy(0)} e^{-t/T_2^*}$	GRE/FID.

Common Signals		
Spin echo	$S_{SE} = \rho (1 - e^{-TR/T_1}) e^{-TE/T_2}$	T1 by TR, T2 by TE.
GRE	$S_{GRE} = \rho \sin \alpha (1 - E_1) / (1 - E_1 \cos \alpha) e^{-TE/T_2^*}$	$E_1 = e^{-TR/T_1}$.
Inversion	$M_z(TI) = M_0 (1 - 2e^{-TI/T_1})$	$TR \gg T_1$.
Null time	$T_{null} = T_1 \ln 2$	Suppress tissue.
Ernst	$\alpha_E = \arccos(e^{-TR/T_1})$	Max steady-state signal.

Weighting Rules		
PD	TR long, TE short	Minimize relaxation weighting.
T1	TR short, TE short	T1 recovery dominates.
T2	TR long, TE long	T2 decay dominates.
T2*	GRE with TE long	Sensitive to dephasing.

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Week 5 - SNR And Hardware

Signal And Noise		
Signal voltage	$U_{sig(x)} = \omega M_{xy(x)} C(x) \Delta V$	Receive sensitivity C.
Noise power	$P = q E ^2 \Delta V$	Sample losses.
Noise variance	$\Psi = 4 k_{BTBW} R$	Johnson-Nyquist.
SNR	$SNR = U_{sig} / U_{noise}$	Image quality.
SNR scaling	$SNR \propto \Delta V / \sqrt{NSA} / \sqrt{BW}$	Resolution/speed tradeoff.

Optimization		
Increase signal	$\uparrow C, \uparrow M_{xy}, \uparrow \omega, \uparrow \Delta V$	More voltage.
Reduce noise	$\downarrow BW, \downarrow T, \downarrow R_{coil}$	Less thermal noise.
Field scaling	$M_0 \propto B_0, \omega \propto B_0$	U_{sig} roughly grows strongly with B0.
Surface coil	$C(x)$ high near coil	High local SNR.

Resolution Limits		
Fourier	$\Delta x \approx \pi / k_{max}$	Sampling aperture.
Relaxation blur	$k^* = \gamma G T_2^*$	Finite T2* filters k-space.
Diffusion blur	$x^2 \approx 6 D T_{acq}$	Long readouts blur.
SoS combine	$I_{SoS} = \sqrt{\sum_c I_c ^2}$	Magnitude coil combine.

Week 6 - Flow Imaging

Motion Phase		
Trajectory	$r(t) = r_{0+} + v(t-t_0) + a(t-t_0)^2/2 + \dots$	Moving spin.
Phase	$\varphi = \gamma \int G(t) r(t) dt$	Gradient phase.
Moments	$M_n = \int G(t) (t-t_0)^{ndt}$	nth gradient moment.
Expansion	$\varphi = \gamma [r_{0M0} + v M1 + a M2/2 + \dots]$	Position/velocity/acceleration.
Bipolar	$M_0 = 0, \varphi \approx \gamma v M_1$	Velocity encoding.

Phase Contrast		
Encoding	$VENC = \pi / (\gamma M_1)$	Phase reaches $\pm \pi$.
Velocity	$v = (\Delta \varphi / \pi) VENC$	Phase difference map.
Aliasing	$ v > VENC \Rightarrow$ phase wraps	Set VENC high enough.
Flow rate	$Q = \sum v \Delta A_i$	Through-plane flow.
Velocity distribution	$s(x, k_v) = \sum p(x, v) e^{jk_{vv}}$	Generalized velocity encoding.

Hemodynamics		
Poiseuille	$Q = \Delta P / R, R \propto 1/d^4$	Diameter dominates.
Reynolds	$Re = \rho v d / \mu$	Laminar if $Re < \sim 2000$.
Shear stress	$SS = -\mu \partial v / \partial r$	Wall shear.
Contrast agent	$1/T_{1,app} = 1/T_{1+R} + 1/c$	Relaxivity model.

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Week 7 - Motion And Artifacts

Motion Encoding		
Phase	$\varphi = \gamma \int G(t)x(t)dt$	All motion artifacts start here.
Motion model	$x(t) = x_0 + v(t-t_0) + a(t-t_0)^2/2 + \dots$	Taylor expansion.
Moment nulling	$M_0 = 0$ removes position phase; $M_1 = 0$ compensates velocity	Gradient design.
Velocity phase	$\varphi_v = \gamma v M_1$	Residual if M1 nonzero.
Velocity spread	signal $\propto \text{sinc}(\beta \gamma A \Delta t \Delta y/2)$	Intravoxel dephasing.

Artifact Rules		
Intra-TR	motion during readout/echo $\Rightarrow$ phase errors and signal loss	Flow, pulsation.
Inter-TR	motion between ky lines $\Rightarrow$ ghosting/blurring	Respiration/patient.
Periodic ghosts	ghost spacing in phase direction $\propto 1/(f_{\text{motionTR}})$	Regular motion.
Random motion	random ky phase $\Rightarrow$ diffuse blur	Irregular motion.

Correction		
Rigid transform	$R(r,t) = A(t)r + d(t)$	Rotation plus translation.
Prospective gradients	$G'(t) = A(t)^{-T}G(t)$	Follow anatomy.
Phase correction	$\varphi'(t) = \varphi(t) - \gamma G(t) \cdot d(t)$	Translation compensation.

Week 8 - fMRI And Diffusion

BOLD/fMRI		
T2*	$1/T_{2^*} = 1/T_{2^*} + \gamma \Delta B$	Static dephasing.
Susceptibility	$\Delta B \approx \Delta \chi B_0$	Blood oxygenation effect.
GRE BOLD	$S(TE) = S_0 e^{-TE/T_{2^*}}$	TE near T2*.
Small change	$\Delta S/S \approx -TE \Delta R_{2^*}$	$R2^* = 1/T2^*$.

Diffusion Physics		
Fick	$j = -D \nabla c(r,t)$	Diffusion flux.
Diffusion eq.	$\partial c / \partial t = D \nabla^2 c$	Conservation plus Fick.
RMS displacement	$R_{\text{rms}} = \sqrt{6D\Delta t}$	3D free diffusion.
Einstein	$D = v^2 \tau / 6$	Random walk.

DWI/DTI		
DWI signal	$S(TE,b) = S_0 e^{-TE/T_2} e^{-bD}$	Scalar diffusion.
b-value	$b = \gamma^2 G^2 \delta^2 (\Delta - \delta/3)$	PGSE sensitivity.
Tensor	$\ln(S/S_0) = -b \ g^T D g$	Direction g.
MD	$MD = (\lambda_1 + \lambda_2 + \lambda_3)/3$	Mean diffusivity.
FA	$FA = \sqrt{(3/2)} \sqrt{\Sigma (\lambda_{i-MD})^2} / \sqrt{\Sigma \lambda_i^2}$	Anisotropy.

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Week 9 - Advanced Imaging

General Encoding		
Data model	$d = E\rho + \eta$	General MRI inverse problem.
Cartesian FT	$E = \text{Fourier sampling operator}$	Standard MRI.
Parallel	E includes coil sensitivities $c_y(r)$	SENSE/arrays.
Optimal inverse	$\hat{\rho} = (E^H \Psi^{-1} E)^{-1} E^H \Psi^{-1} d$	If well-conditioned.
Regularized	$\hat{\rho} = (E^H \Psi^{-1} E + \lambda I)^{-1} E^H \Psi^{-1} d$	For high R.

Compressed Sensing		
Sparsity	$x = \Phi\rho$ has many small/zero coefficients	Sparse transform.
Undersampled data	$d_\Omega = P_\Omega \Phi\rho$	Sample subset Ω .
CS recon	$\min_\rho \ \Phi\rho\ _1$ s.t. $\ E\rho - d\ _2 \leq \epsilon$	Sparse recovery.
TV	$\Phi\rho = \nabla\rho \Rightarrow$ total variation penalty	Piecewise smooth images.
Sampling	random/incoherent k-space undersampling	Artifacts become noise-like.

Limits		
PSF	$\text{PSF}(r) = \text{FT}^{-1}\{\text{sampling pattern}\}$	Undersampling artifact shape.
Acceleration	higher R needs sparsity + SNR + calibration	No free lunch.
Low-rank/dynamics	suppp small or temporal basis small	Advanced priors.

Week 10 - Spectroscopy I

Chemical Shift		
Larmor	$\nu = \gamma B_0$ [Hz]	Sign by convention.
Shift	$\delta = (\nu - \nu_{\text{ref}}) / \nu_{\text{ref}} \cdot 10^6$ ppm	Field-independent ppm.
Hz separation	$\Delta\nu = \Delta\delta \cdot \nu_{\text{ref}} \cdot 10^{-6}$	Grows with B0.
1 ppm	1 ppm = $\nu_{\text{ref}} \cdot 10^{-6}$ Hz	64 Hz at 1.5T for ^1H .

Spectrum		
FID	$s(t) = A e^{-t/T_2^*} e^{i2\pi\nu_{\text{At}} t}$	Single component.
Spectrum	$S(\nu) = \text{FT}\{s(t)\}$	Frequency-domain signal.
Lorentzian	$\text{Re } S(\nu) \propto T_2^* / [1 + (2\pi T_2^* (\nu - \nu_{\text{A}}))^2]$	Line shape.
Linewidth	$\text{FWHM} = 1/(\pi T_2^*)$	Shorter T2* broader.
Mixture	$s(t) = \sum_{\text{cwe}} e^{-t/T_2^*} e^{i2\pi\nu_{\text{ct}} t}$	Superposition.

Sampling And J		
Acquisition	$T_{\text{acq}} = Ndt$	N samples.
Bandwidth	$\text{BW} = 1/dt$	Hz.
Resolution	$d\nu = 1/T_{\text{acq}}$	Peak spacing.
Resolve peaks	$d\nu \leq \Delta\nu/2$	Rule of thumb.
J multiplet	N coupled spin-1 nuclei $\Rightarrow 2NI + 1$ lines	Binomial intensities.
J echo	$\text{TE} = n/J$	Odd n flips doublet sign.
Editing	$\text{TE} = 1/J$	Difference editing.

Week 11 - Spectroscopy II

Suppression And Inversion		
Water/fat	$\Delta\delta\approx 3.5\text{ ppm}$	Water 4.7, fat 1.2 ppm.
Phase accrual	$\Delta\phi=2\pi\Delta\nu\tau$	Chemical shift phase.
Binomial spacing	$\tau\approx 1/(2\Delta\nu)$ or $1/(4\Delta\nu)$	Depends on target phase.
Inversion recovery	$M_{z(T)}=M_0(1-2e^{-T/T_1})$	After 180°.
Null	$T_{null}=T_1\ln 2$	Water/fat suppression.

Localization		
Surface coil	$S(r)=B_{1(r)}\sin[\gamma B_{1(r)}t]$	Sensitivity + flip angle.
Adiabatic	$ d\theta/dt \ll \gamma B_{eff} $	Robust inversion/excitation.
PRESS/STEAM	Echo at TE; stimulated echo stores Mz during TM	Voxel selection.
Phase cycling	linear combinations isolate voxel term	Remove unwanted echoes.

CSI And High Field		
Shift artifact	$\Delta x=\Delta\nu_{CS}/\gamma G$	Chemical shift displacement.
Fractional shift	$\Delta x/\Delta x_{RF}=\Delta\nu_{CS}/BW_{RF}$	Use large RF BW.
CSI signal	$s(k_x,k_y,t)=\iiint \rho(x,y,\Delta\nu)e^{ik_{xx}x+ik_{yy}y+i2\pi\Delta\nu t}dx dy d\Delta\nu$	2 spatial + spectral.
CSI recon	$S(x,y,\Delta\nu)=FT_{tFTk_xFTk_y}\{s\}$	3D transform.
CSI scan time	$T_{scan}=N_xN_yT_R\cdot NSA$	Slow phase encoding.
High-field SNR	$U_{sig}\sim B_0^2, U_{noise}\sim B_0, SNR\sim B_0$	Approximate.
SAR	$P\propto\sigma E^2\propto\sigma V_0^2B_0^{-1}$	High-field cost.